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卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-CS-0508 30-B003701S-T0508

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图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

结构 专业 第 05 卷 第 08 册
Structure SPECIALITY VOLUME BOOK

卷册名称 工业、消防水池连通结构图
Volume name Drawings of industrial and fire pool connection structure

批 准 APPROVED BY	2019.09.26
审 核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	刘康平 2019.09.26
图 纸 2 张 DRAWING NOS.	打印件 0 本 PRINTED COPIES NOS.

序号 No.	图 号 DRAWING NO.	文 件 名 称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	30-B003701S-T0508-001	工业、消防水池连通结构平面布置图 Plan of industrial and fire pool connection structure	A	1		
2	30-B003701S-T0508-002	应急补水泵组基础及电缆井详图 The details of emergency replenishment pump base and cable well	A	1		

注:

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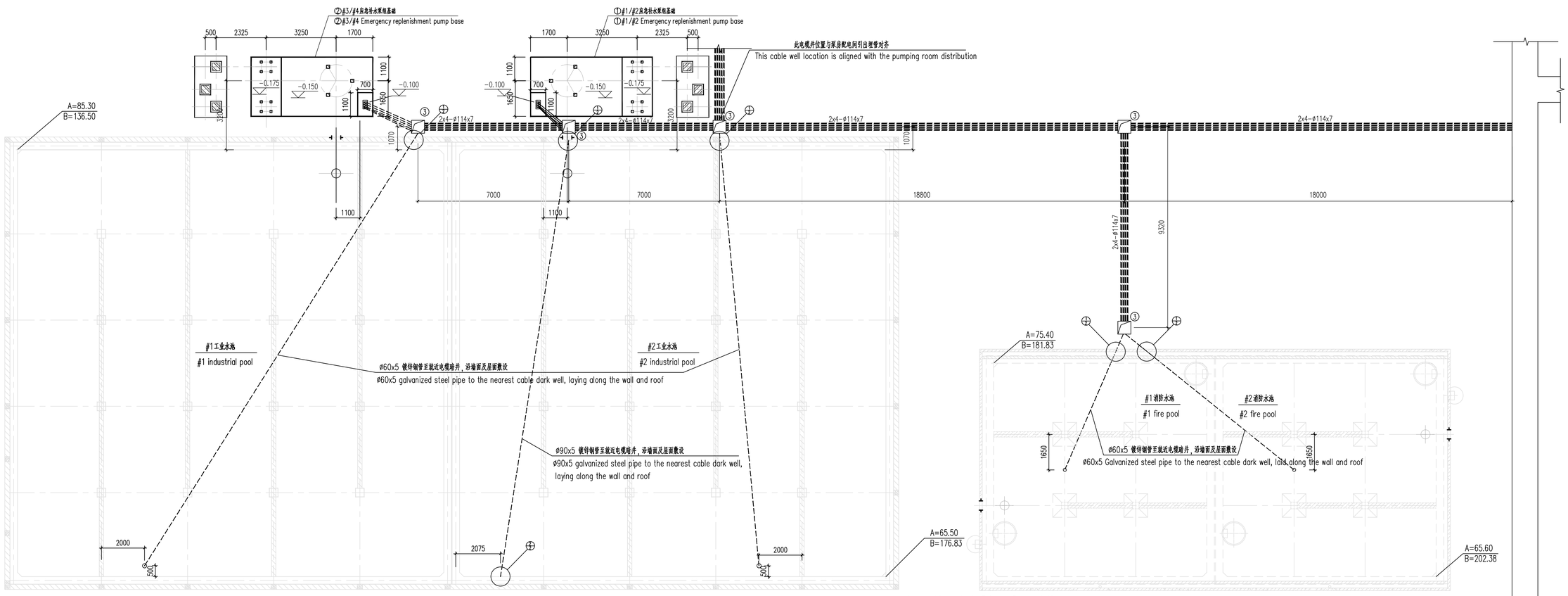
基础一览表

编号	基础名称	单位	数量	基础大小 (m x m)	基础方量(m^3)(含垫层及二次灌浆) 单件	小计	基础施工图号	备注
①	#1/#2应急补水泵组基础	只	1		16.41	16.41	30-B003701S-T0508-002	
②	#3/#4应急补水泵组基础	只	1		16.41	16.41	30-B003701S-T0508-002	
③	电缆井	只	5	0.9X0.9	0.52	2.60	30-B003701S-T0508-002	(不包括盖板)
总计	混凝土总量:41.60m^3							

The list of fundations

No.	Name	Quantity	Size (m x m)	Volume (m^3) (Including bedding and secondary grouting) Single	Subtotal	Drawing number	Remarks
①	#1/#2Emergency replenishment pump base	1		19.50	19.50	30-B003701S-T0508-002	
②	#3/#4Emergency replenishment pump base	1		19.50	19.50	30-B003701S-T0508-002	
③	cable well	5	0.9X0.9	0.52	2.60	30-B003701S-T0508-002	(excluding the cover)
Total	Total amount of concrete: 41.60m^3						



工业、消防水池连通结构平面布置图

Plan of Industrial and fire pool connection structure

说明:

1、场地-0.30相当于绝对标高193.40(EGM2008高程系统)。如无特别说明，本工程标高均采用相对标高。
2、采用巴基斯坦UTM-43N平面坐标系，相对平面坐标以米(m)为单位，尺寸以毫米(mm)为单位。
3、应急补水泵组基础基底以下宽度沿底板边护500mm范围内用级配碎石换填持力层，
基础的持力层为①粉质黏土,地基承载力特征值为70kPa。
4、混凝土技术要求:(1)水胶比不大于0.50，(2)氯离子含量不大于0.1%，
(3)水泥型号应满足ACI-318-14表19.3.2.1中暴露类别S1对应水泥
(Cementitious materials-Types)的要求，(4)抗渗等级0.8MPa，(5)掺矿物掺合料。
电缆井及设备基础fc'=28MPa;二次灌浆层强度等级:fc'=35Mpa;混凝土垫层fc'=21MPa。
钢筋技术要求:应满足ASTM A706, Grade 60[420]变形钢筋
或满足ACI 318-14第20.2.2.5节要求的ASTM A615, Grade 60[420]变形钢筋。
钢筋的抗拉强度实测值与屈服强度实测值的比值不应小于1.25;钢筋的屈服强度实测值与
屈服强度标准值的比值不应大于1.3，且钢筋在最大拉力下的总伸长率实测值不应小于9%。
Φ代表Grade 40，Φ表示Grade 60 钢，钢材Q235B，电焊条E43XX，焊缝高度Hf=hmin。
5、受力钢筋的混凝土保护层厚度:设备基础底为50mm。
6、所有外露铁件均热镀锌防腐，焊饰处防锈漆一道，银粉漆二道。
7、本工程结构施工时应与水、电等其他工种图纸配合施工。
8、施工区域下有前期埋设的电缆、水管,施工时应注意避免挖断。
9、埋管管顶标高-0.80m，埋管转弯半径不小于8d,各管口伸出基础面50mm，每根弯头不超过3个，直角弯不超过2个。
10、外露基础四周外缘设置R=30mm的倒角。
11、其它未尽事宜请严格按照有关的规程规范、标准图集等的相关内容执行。
12、本卷册需结合T0501、T0502卷册施工。

Specification:

1、The site level of -0.300 is equal to the absolute elevation of193.40(geodestic datum system EGM2008). If not specified,the elevations of the project are adopt to the relative elevation.
2、Using Pakistan UTM-43N plane coordinate system,the relative plane coordinates are in meters (m) and the dimensions are in millimeters (mm).
3、The width of the foundation of the emergency replenishing pump group is filled with graded gravel and the bearing layer within 500mm of the bottom of the bottom plate.
The foundation bearing layer is silty clay①, and the bearing capacity characteristic value of the foundation is 70 kPa.
4、Concrete technical requirements: (1)the water binder ratio is not greater than 0.50,
(2)the content of chloride ion is not more than 0.1%, (3)the cement shall meet the requirements of the exposure category S1 in the Table 19.3.2.1 of ACI-318-14, corresponding to the cement (Cementitious, materials-Types),
(4)the impermeability grade is 0.8MPa, (5)the mineral admixture is added.
For foundations: fc'=28Mpa,For grounting: fc'=35Mpa,For concrete cushion: fc'=21Mpa.
Reinforcement technical requirements: (1)It shall meet the requirements of ASTM A706,
Grade 60[420] deformed steel bars or meeting the requirements of section ACI 20.2.2.5 318-14.The ratio between measured tensile strength and measured value of yield strength should not be less than 1.25;The ratio between the measured yield strength and the standard value of yield strength should not exceed 1.3; The measured value of the total elongation at the maximum tensile force should not be less than 9%.
Reinforcement Φ: ASTM A615 Grade 60, fy = 420MPa.,Reinforcement Φ: ASTM A615 Grade 40,,Steel: Q235B, welding rod: E43XX, weld height: Hf=h_{min}.
5、The thickness of the protective layer of the reinforcement: The bottom of the foundation is 50mm。
6、All exposed iron parts are hot-dip galvanized anti-corrosion,welding wound apply a antirust paint and two silver powder.
7、The drawings should be checked with drawings of other specialty before construction.
8、There are cables and water pipes buried in the construction area in the early stage, and be careful to avoid cutting during construction.
9、The turning radius of the buried pipe is not less than 8d, and each nozzle extends 50mm from the base surface.
Each steel pipe is set not more than 3 elbow and not more than 2 right angle bends.
10、External around of the foundation above ground should be filleted with the fillet radius 30mm.
11、For other unfinished matters, please strictly follow the relevant regulations, standard atlas and other related content.
12、This drawings must be combined with T0501 and T0502.

专业 Speciality	会签 Sign	日期 Date
电气一次	郑煜	2019.09.26
水工	甲瑞	2019.09.26

Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程
±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站
±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 5 0 8 - 0 0 1 A

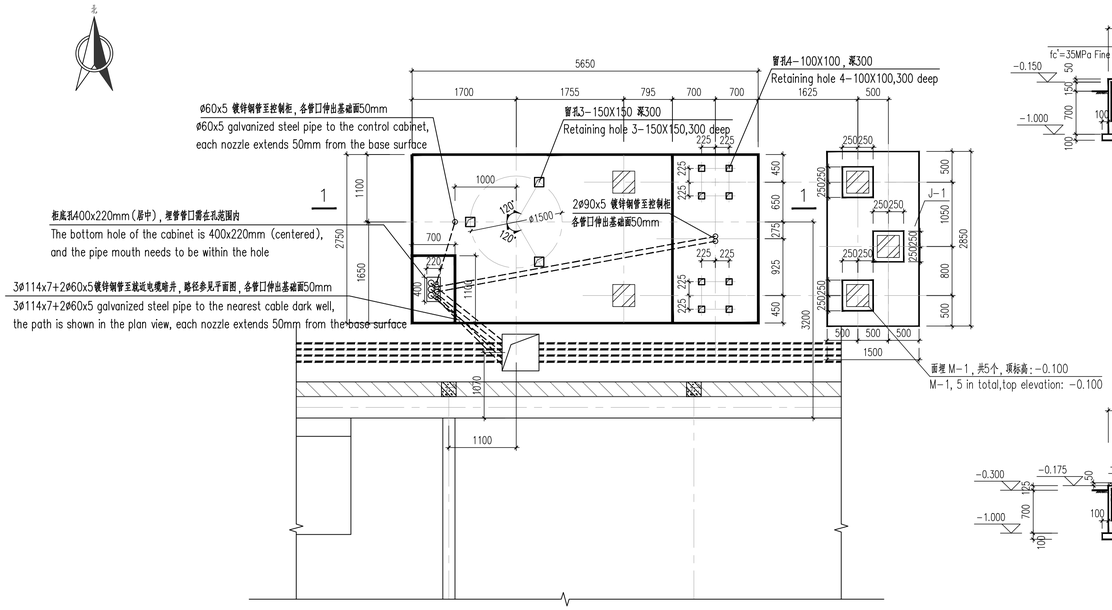
	姓名 Name	日期 Date
设计 Designed	刘庆平	2019.09.26
校核 Checked	魏伟	2019.09.26
审核 Reviewed	孙雪婧	2019.09.26
批准 Approved	杨飞	2019.09.26
比例 Scale	1:100	图号 DWG NO.

工业、消防水池连通结构平面布置图
Plan of industrial and fire pool connection structure

30-B003701S-T0508-001

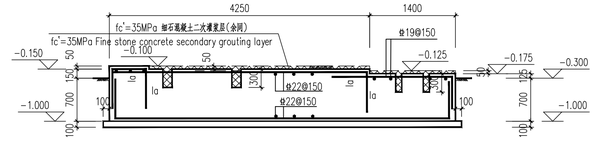
版本 REV. A

A1(841x594)

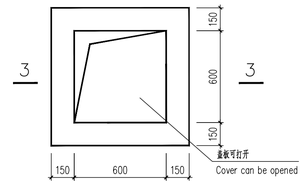


#1/#2应急补水泵组基础平面布置图

Plan of #1/#2 Emergency replenishment pump base

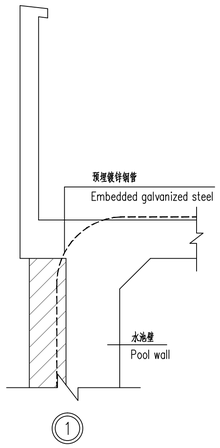


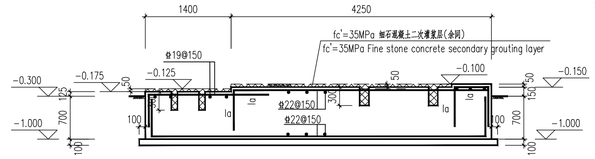
1-1



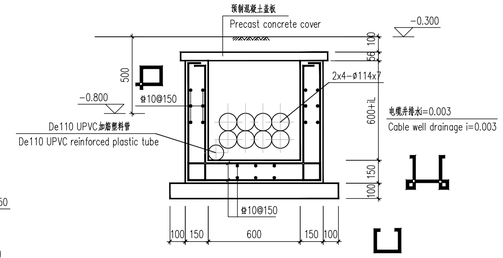
电缆井

Cable well





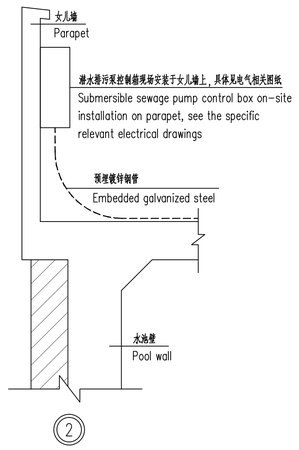
2-2

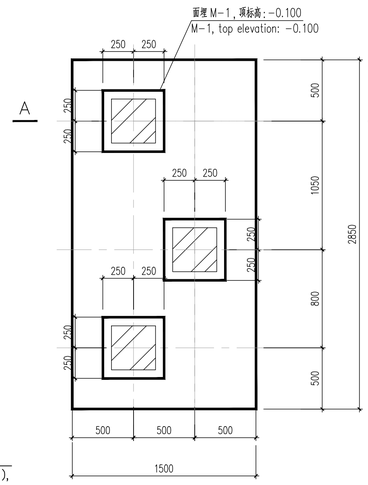


3-3

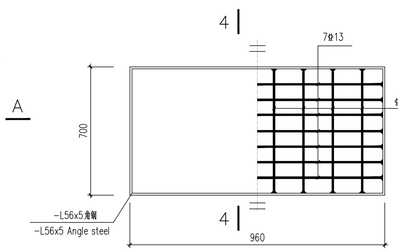
注：盖板四周设橡胶垫圈

Note: Rubber gasket is placed around the cover



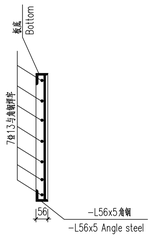


J-1

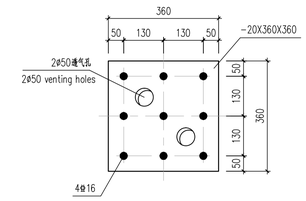


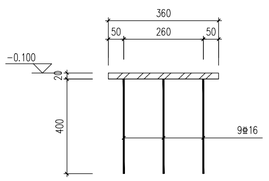
预制混凝土盖板

Precast concrete cover



4-4

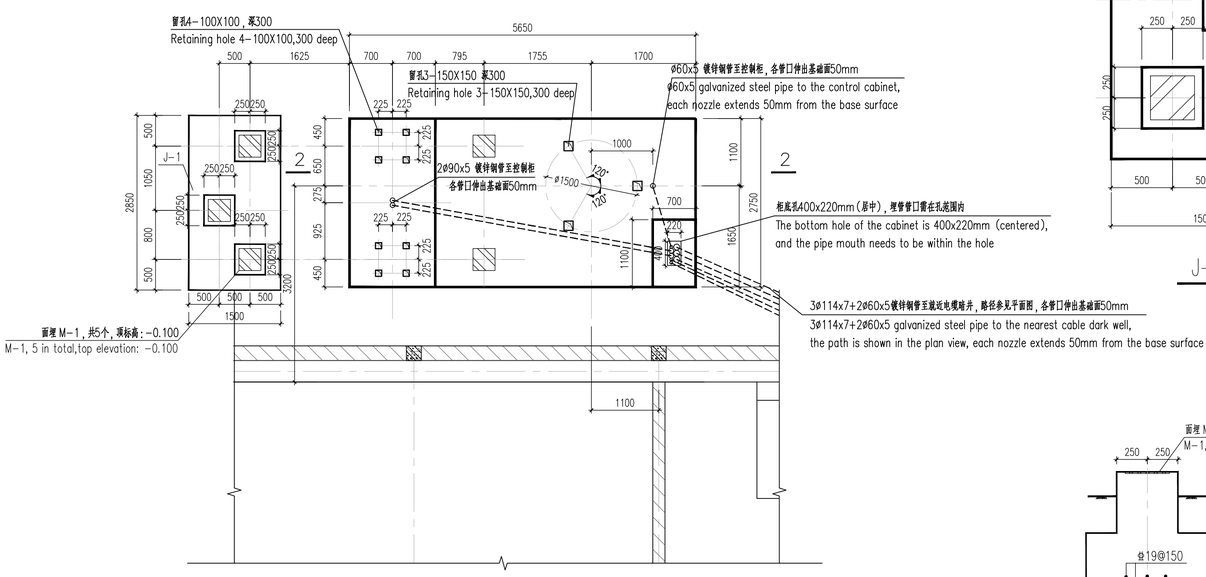




M-1

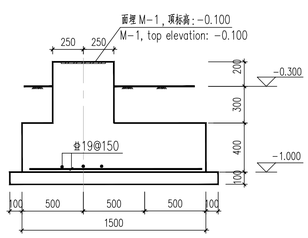
注：钢筋与钢板间采用压力埋弧焊

Note: Pressure submerged arc welding is used between steel bars and steel plates.



#3/#4应急补水泵组基础平面布置图

Plan of #3/#4 Emergency replenishment pump base



A-A

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

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±660kV HVDC Project from Matiari to Lahore in Pakistan

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拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE：施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 5 0 8 - 0 0 2 A

	姓名 Name	日期 Date
设计 Designed	刘康平	2019.09.26
校核 Checked	魏祐	2019.09.26
审核 Reviewed	舒清清	2019.09.26
批准 Approved	杨必	2019.09.26
比例 Scale	1:100	图号 DWG NO.

应急补水泵组基础及电缆井详图

The details of emergency replenishment pump base and cable well

30-B003701S-T0508-002

版本 REV. A

专业 Speciality	会签 Sign	日期 Date
电气一次	郑煜	2019.09.26
水工	甲瑶	2019.09.26

A1(841x594)